

ActInf Livestream #007.2: "Variational ecology and the physics of sentient systems" (2019)

Source: [YouTube](#) Playlist: Active Inference Livestreams (Paper discussions)

Duration: 1:12:36 | Uploaded: Unknown Transcript method: auto_caption

hello and welcome everyone to the active inference live stream this is active inference live stream 7.2 it is november third twenty twenty welcome to team com everyone we are an experiment in online team communication learning and practice related to active inference you can find us on twitter at inferenceactive at email our public key base team or on youtube this is a recorded and an archived live stream so please provide us feedback so that we can improve on our work all backgrounds and perspectives are welcome here and also for video etiquette on live stream remember to mute if it's noisy raise your hands so we can hear from everyone in the stack and use respectful speech today in act imp stream 7.2 we are going to go through our introductions and have some warm-up questions and then we'll get into the discussion of 7.2 we'll talk about the paper variational ecology and the physics of sentient systems ramstead it all 2019 and we'll go over the goals and the abstract and the roadmap then we'll talk about some big ideas and questions in ecology and then we'll go through the figures i remember last time we mentioned figure six and so we've gone the figures in reverse order so we'll start with six and we'll just go down from there and also for the rest of 2020 we will be discussing papers 8 9 10 and 11 which we'll go through in a second and just check out our twitter or message us if you have any questions so what are papers 8 9 10 and 11 and the dates paper 8 scaling active inference on november 10th and 17th all discussions are at uh 7 30 to 9 00 pst pacific standard time in the morning paper 9 is projective consciousness model and phenomenal selfhood paper 10 is a variational approach to scripts and paper 11 is sophisticated active inference which as we were just talking about before the stream we'll also have some cool accessory content let's go to our intros and warm-ups so for this part just introduce yourself and your location you can say hello and give a short introduction and then pass to someone who hasn't spoken yet so i'm daniel i'm in davis california and i'll pass it to shannon hi guys i'm shannon i'm usually in merced california but i'm in south dakota for the pandemic and i'm going to pass it to ivan hi everyone my name is ivan i'm from moscow

russia and i pass it to sasha hi everyone um i'm sasha and i'm in california and i'll pass it to marcus hi i'm marcus um i am in lean shopping which is a small town in sweden a couple of hours outside of stockholm um and i gave a brief introduction to myself because i'm new here so my i did my phd in medical imaging and my research is mainly focusing on cardiovascular mri so quite different from fap and active inference which has become sort of a hobby for me the last couple years and i find these live streams very good to explain these concepts and i'm looking forward to the discussion today so i'm gonna pass over to cameron hi i'm cameron i'm in zurich i was here for the first time last week and i just started a phd in philosophy in zurich uh and i passed it on to alex i'm alex i'm in moscow russia i'm a researcher in system management school and i pass it to maxwell right thank you hello everyone uh my name is maxwell ramstead i'm a postdoctoral fellow at mcgill university and i'm zooming in from just outside montreal in the suburbs and i'm a first author on this paper so very excited to discuss it with you i i'm not has everyone spoken yet i i lost track okay great thanks for all the introductions let's go to the warm-up questions and here it'd be cool to hear from everyone and you can just answer as narrowly or as broadly as you like and go ahead and raise your hand if you'd like to speak here the first question is what is your niche or what could it be so i'll just start with my local niche right now i'm looking at the video screens i'm trying to balance getting the information from people's face versus the slides um what's another person's just niche right now around them or a related question is what skills help us find affordances in a changing world so our niche definitely presents us with uncertainty multiple scales perhaps of uncertainty what's behind me in the door or what happens in the deep future so how do we find affordances or how do organisms find affordances in that kind of a context i liked your mention of your niche is the computer screen and looking at people's faces i feel like for every different communication program like if it's zoom or jitsi we can organize our digital environment different ways so we can like see different people's faces or see different aspects of the screen to allow for as much non-verbal communication as we want or need to make that particular communication effective cameron um for the first question um going back to a niche i constructed for myself about a year ago when i was coaching tennis would often work in the late afternoon to late in the evening and i noticed how my diet changed to make sure i had enough high enough sugar level so that i could give really energetic lessons to the kids in the late afternoon and then sort of tone it down in terms of an energy level for the adults and then uh have a massive kind of slump as soon as i got home so i could sleep properly for the next kind of round of coaching so i noticed how my kind of diet set up my energy levels just for

that kind of lifestyle and as soon as i stopped coaching tennis it was uh away with the biscuits and chocolate yep that kind of just reminds me of like looking forward to dinner you're not exactly sure what it's going to be but it's going to be a physiological thing it's going to reduce your uncertainty about food anyways what is one thing that you are curious about or you want to explore in today's discussion so that could be something that you wanted to figure out from the paper or something that the paper kind of resonated with you or just something from the side from the rest of your life that came into context today when thinking about this paper so i'll start one thing that kind of motivated me in the slides was just exploring how the variational ecology is linked to big questions in so-called i guess traditional ecology non-fep non-active inference ecology how do we make that connection because if that connection can be made then the expansion of the theory from just a few equations to something that's in use in a lot of applied situations would be a lot more easy any other random thoughts or that's totally chill we can just go on to the paper and people can raise their thoughts as they see fit i mean i'd be curious to hear uh like everyone else on this like what did you hope to get from this uh because as the author i can respond to some of these questions so let's go cameron and then shannon muted sorry at the very end uh maxwell you you write about shared intentionality and i wondered how you think about the debate between shared intentionality and theories of mind and that kind of kind of very mentalistic idea of interaction yeah i had that sort of same topic of shared intentionality i was also wondering um in our universities like seminary yesterday we were talking about collective memory and i wonder whether this like variational ecology can tell us anything about what we choose to remember or what we end up remembering as a culture or as a society about our culture's past it might not it might be that might be a little abstract for what we're talking about currently since it's not necessarily behavior or it's a behavior that we have to verbalize and speak out loud but those are great questions yep i agree on the part about remembering it's like there's this multi-level remembering process how much data do we retain from yesterday from two seconds ago from 200 years ago from 2 million years ago and then there's all always different perspectives on just the facts or the events of whatever did or didn't occur and all this narrative weaving that occurs so how we model and remodel how we update our models of those multi-scale memory processes some of them are extended some are internal that's everything so i think it's really interesting question i hope we'll return to that sasha um yeah that's a kind of great point to link it all together that um individuals that are interacting um can spur more i guess memories or common experiences by revealing more or less about their current niche um and so yeah just

really thinking about how individuals should um or should not interact um when trying to build common ground um and how much of their niche they should be sharing uh in that communication but yeah a lot of uh fun things with that in the virtual space um that we can reveal to each other cool i think that hopefully sets the stage to talk about some of the specifics of the paper and also big ideas in ecology and also things are happening all around us so the paper is variational ecology in the physics of sentient systems from 2019 and we reviewed the goal of the paper last week but the shorthand kind of equation form is that variational neuroethology vne which is about organisms and their neurobiology primarily can be kind of jointed together with vanc the variational approach to niche construction and then that summation that continuation of explanation from kind of the neurobiology into the extended niche is called variational ecology and then the non-active inference phrasing would be something like how do you make a multi-scale ecology framework that builds on ecology evolution and development and also adds in insight from physics mathematics and complexity we also talked about the roadmap last week and sort of formulaic in that it introduces the two separate ideas in section two and three and then adds them together to result in a synthesis framework represented here and with the connector added between the uh active orange connector to the purple so to kind of just start the discussion i wanted to begin with some big questions in ecology and some other people may have done ecology in a specific formal or informal way but there's so much to it and it's just interesting to connect it to the questions that motivated that field before this kind of synthesis of even ecology evolution and development which is its own pyramid of you know ideas that connected and people who bridged evolution and ecology and all these kinds of things so let's just look at the big questions in ecology and think about the context that this synthesis is happening in and then think about all the cases where ecology is applied so conservation and policy planning and stuff and then think about how a mapping between fep and active inference broadly and could allow for a lot of translational and applied use so here's one kind of interesting historical coincidence it's a 1983 paper by richard lewinton called the organism as the subject and object of evolution so this is kind of before the whole uh physics in biology approach was becoming mainstreamed but he was coming from a very integrationist perspective and he wrote before darwin theories of historical change were all transformational that is systems were seen as undergoing change in time because each element in the system underwent an individual transformation during its life history so like the muscle cells became different or lamarck's theory of evolution was transformational in regarding species as changing because each individual organism

underwent the same change through inner will and striving an organism would change its nature and that change would be transmitted to offspring so that's what people often call the marquee in evolution now here's where the historical coincidence is in contrast darwin proposed a variational principle now it turns out that this is a different variational principle than the variational principle from physics that we're going to be discussing but i thought it was really interesting that lewinton and darwin use the same idea and so what's the idea individual members of the ensemble differ from each other in some properties and that the system evolves by changes in the proportion of different types there's a sorting out process dot dot and so variation among objects is transformed into temporal variation a dynamic process arises in time as the consequence of static variation in space there's no historical process other than the evolution of living organisms that has this variational form at least as far as we know so that's actually going a long ways towards this sort of what is schrodinger's question about what is life just an interesting way that variational has interwoven from the history of biology from darwin from lewinton and now finding itself with a formal definition ironically with the same name related to variational principles so anyways sorry for the little bit of a uh history side let's get to the ecology questions and if anyone has thoughts on that previous link they can kind of address it here these are some questions drawn from a paper by sir robert may called unanswered questions in ecology so we can figure that they're probably still unanswered but let's just look at some of these and if anyone wants to reflect on it how do they see it playing out or how do they think or wonder that active inference could play into it i'll put up the first two what determines population density and what determines the spatial structure of population these are actually like the first two that may brings up so what what could an answer to this look like uh or where do you think the fep and ecology could intersect here i'll just list them all also how can we study stability and complexity in ecological communities and what is the proper scale of analysis for ecological studies in space and time so what might be an approach or what might be a mapping that we could draw here one thought is the proper scale of analysis for ecological studies is kind of related to our discussion about what time scales ergotic systems or the ergotic hypothesis exists at because there might be some time scales or spatial scales that don't make sense to study like if you're studying the lion hunting the gazelle the centimeter squared time scale or spatial scale in the one second might not capture some important dynamics of the system but then if you study it at too broad a level it might be not able to be studied for another way just happening too fast underlying so just one thought there anyone wanna yeah shannon go ahead yeah so i don't study ecology but um i

know we've learned some about like predator prey models so population density is determined by you know the amount of resources in the environment and as the prey depleted then the predators depleted because they don't have enough food so the prey can come back and you have this structure over time and the same kind of thing happens even with trees and these beetles that eat trees so you'll see like forests even over a much longer time scale having um like trees that die out and then they grow back eventually and the beetle population will also follow that predator prey structure and die out and come back yes so there's these rhythms and some slower ones some are across species some are within a species they're all related to measurements so at the very least we're going to be able to use different mathematical frameworks to analyze our measurement just like we use different statistical tests but the deeper question is whether the free energy principle or whether the active inference framework describes what these ecological communities are doing or what is actually occurring versus to what extent it's a really convenient model fitting approach um all right the second next slide yeah what would the difference in implication be i mean uh i think uh the proper philosophy of science attitude to have with regards to any scientific model whatsoever is that truth has nothing to do with what science is doing i mean this might be slightly controversial but i think science has to do with uh model comparison effectively i mean all of science has moved into this kind of kind of game i think where what we're doing is so first of all data is never presented in a naked way right uh so you know your neuroimaging data for example will be organized as a kind of a heat map of the brain for example like data is always presented as a model of data and basically what we're doing is confronting different theoretical and formal models to find which explains away the variance in our data the best so from that point of view we needn't read the free energy principle as making any literal claim about anything just the same way that you know uh any other scientific model is just going to be kind of a it's a useful description so i'm i'm putting on my mel andrews hat today and being an anti-realist but uh or i mean a non-realist right like it's just uh it's not like newton's laws became false after you know they were shown to be a special case of relativistic uh you know dynamics these are just like successively better models that successively and iteratively explain away the variance in our data better but i don't know if it's appropriate to ask the question you know are these merely models or actually the way that the world works i mean i mean that's the quote right all models are wrong but some are useful exactly yep and then the what is useful must always be unpacked to to whom and who else is influenced by it and then what is their value structure and all these types of second order effects and and i agree that if nothing else it's a useful

multi-scale framework just like a multi-level regressor regressor model or the gaussian distribution i mean linear modeling is used for some computational benefits so yeah it's unfortunate when it does constrain the phenotypic distributions in the world but perhaps with flexible tools we could actually accommodate for very very high dimensional preference systems or things like that rather than using like a excel spreadsheet that takes an average and then aggregates that into the next level of the aggregator so how could we go and look at variation across multiple levels of the system and yeah good interesting things there let's look at a few more just classic ecology questions before we go to the figures so this is from a 2013 paper in journal of ecology called identification of 100 fundamental ecological questions so those are just the little small ones um i'm just going to list all 20 of them that i just grabbed the first 20 um so people could can uh think just you know pick a random number spin something roll roll three die whatever it is but these were just the questions that this large set of authors basically used some polling and discussion methodology to converge on these 100 questions and so they definitely thought about which ones to put at the beginning and they definitely thought about which ones to include so i just think these are kind of what ecologists broadly may find to be some of the biggest questions so that means that if somebody can make a connection where something that was previously modeled in an incomplete or in a really hetero heterogeneous framework if there's a way to model it in a more unified way then that's something that addresses a fundamental question in ecology and so that's one of the ways that we can apply theory and then think about what are the downstream applications like look at their number one question what are the evolutionary consequences of species becoming less connected through fragmentation or more connected through globalization so that's roads being built that break uh migration corridors that's also barnacles being brought around by global fishing fleets and so different species and different ecosystems are connected in new ways their markov blankets their ergotic levels of analysis these are all changing and so how is that going to shape phenotypic and behavioral and genetic diversity that's conservation ecology so if that can be re-understood or mapped into a active inference framework it's again at the very least a super useful way to think about multi-scale because it includes that bottom-up and top-down element which a lot of these questions implicitly include but often through very constrained ways uh specific types of partial differential equations or specific types of simulations that aren't really grounded in this type of and then um yeah i just think that's kind of an interesting thing so i thought that's that's why it's worth taking the time to at least look through some of these anyone can raise their hand if they want to jump in on one of these 1

through 20 but like it's just useful to think about these because these are the questions that people want to see answered in ecology shannon i mean number five is exactly what we're addressing if we're asking what can multi-scale uh free energy tell us about these different levels of of selection and also different levels of behavior or different levels of insert phenomena you're interested in here yep what are the relative contributions of different levels of selection gene individual group it's a life history evolution and the resulting population dynamics yep definitely something that's super interesting to think about and does a variational principle that links these levels of analysis like if you have a multi-level regression model you don't say that one level has causal priority over the other level it's just a different way that variance was partitioned and that reminds me of evelyn fox keller's metaphor of the two people who are pumping water together and that's the nature nurture so-called and that's how you can uh really do a partitioning of labor only for certain tasks but not for kind of shared tasks like her water pumping example and so maybe the partitioning is a statistical post hoc and some traits will partition highly on to heritability or some traits partition highly onto some other uh uni-dimensional axes but then in a fuller variational framework we would have a more nuanced way to talk about how those parts are related yep and then another question that is definitely related to the whole area is 16. how do organisms make movement decisions in relation to dispersal migration foraging or mate search so definitely a lot of work on everything from the ocular motor epistemic foraging the movement of the eyes towards informative areas of the visual space to focus on to search behavior and not could be in an abstract computational space so many of the optimization algorithms we know are named after animals because they're kind of bio-inspired ant colony optimization cuckoo search all these kinds of things are related to ecologically observed strategies so maybe if we had a way to frame those different strategies in nature we could do bio discovery and then we could implement a lot of cool ways to who knows what you know optimize things make uh swarms of robots that are useful who knows what should happen okay um number eleven also uh that what is the what are the evolutionary and ecological mechanisms that govern species range margins um that reminds me of this uh concept in ecology of uh the fundamental versus realized niche um and so i think that has a lot of uh nice parallels to um kind of how humans behave but also thinking about um cognition and how we uh choose to think about the world and what sort of niches that uh that model is tailored to what is the fundamental versus the um uh full disclosure i also do not study ecology but i'm teaching an intro to bio class to freshmen um and so i'm learning about this uh as i go as well it's it's kind of fun um but the fundamental

is the environment uh the combination of bio abiotic factors that allow an organism to survive so anywhere where you could live um and then the realized niche is uh where you actually do live based on your interactions with other species and other biotic factors that influence your niche do you know what this overlaps with corolla thoughts's concepts of the developmental versus the selective niche so as thoughts basically argues that effectively uh you can think of the niche as the set of selection pressures to which organisms are subject but also as the set of reliably inherited resources that are necessary for the reconstitution of the life cycle in every generation and that in ideally they coincide but they don't necessarily uh so that you know your developmental ideally prepares you for the pressures that you'll be you know undergoing in your selective niche uh but you know for various you know evolutionary reasons or other you might be tuned to the wrong environment for example and then die off so uh but it does seem to overlap with the distinction that you're mentioning here wow very interesting and the changes in the uh in different things can lead to the realized niche where things actually exist if the population hits a critical point then it can just extinct locally or or even regionally and that doesn't that's like a non-uh recoverable state and so it can change uh in a very like the niche can be modified rapidly and so that definitely relates to it let's actually do one more slide on niche and affordances and then this is before we just get to the figure so if anyone has any more like general thoughts now we're bridging the gap we're seeing how the regular ecology brings over to this new way of thinking about it so niche and affordance are linked and that's kind of cool that we just were talking about that because there's the niche example here is the different birds so flamingo is digging the deepest into the dirt to or under the water to get food the um birds on the right are going more shallowly and so their beaks are different their behavior is different everything is different about their biology and they may or may not be able to have uh general versus specialty like maybe the flamingo can also get the shallow but not vice versa or something like that so that's a sort of physical niche concept but our niche includes objects includes other people and trees and bike paths and digital experiences as well so what happens now that technology is part of the human informational and ecological niche maybe bringing in what was just being discussed about the fundamental and the realized as well as the selective and the developmental niche concepts so how could active inference be used to to understand and model and design in these situations so i gotta start there is for politics or for information systems especially with digital there is certainly a vast number of uh degrees of freedom in the design and the realized info niche that we're in is just like literally whichever whichever one we're in however it ends up whoever's behind the scenes

doing whatever that's the realized one we're gonna get but what is the fundamental info niche what is the actual bigger habitable space and which parts of that habitable space might have some features or some benefits that we don't realize through our current uh existence in that realizable so conceptually i would want to distinguish between the physical environment and the uh so the way that we're defining it at least the niche is the set of affordances to which organisms are sensitive so that doesn't overlap exactly with the physical environment um i mean two species can share the same physical environment but not occupy the same niche because they're not sensitive to the same things uh so you know like something may be like a prey item for one and just be like you know part of the background uh for another species uh so i think that's how you get around that uh i was confronted by kim sterily uh when we we talked we gave a talk in australia i think in 2017 or 18 about this and i think this helps to dissolve that issue there isn't like one ecological niche there's one physical environment that we all share and then relative to uh essentially the the skills and the concerns that we embody uh the world will appear very differently and so we we live in different niches even if the physical environment overlaps okay yvonne yeah thank you if if we return that now we share one virtual niche uh we do it by uh our brain work and before the internet we can uh speak with each other just to physically present near or by telephone but now we can we can do it we can choose what type of communication we can choose the type of communication and so we construct different ways to create niche itself yes thanks for sharing that and it's uh i i hear you max well on the niche concept definitely an idea that ecologists have debated over as much as um species but the uh idea of the shared informational resource let's go with that certainly there are things that are becoming knowledge resources that are not only uh non-evenly shared across the globe but also are related to new kinds of informational connections that haven't occurred in any individual sensory niche hearing languages that one wouldn't have heard access to perspectives and the high-throughput nature of media so these are all things that i think change our our match or mismatch or our fitness between our evolutionary priors on biofeedback or on safety or on the veracity of language these kinds of are being connected and being played out in new ways so i would think how can active inference be used to understand and design these situations well we can define the remote team and we can also think of it as almost a reduced case and a known mismatch case with our own like quote natural ecology but first our ecology is so flexible and humans have displayed so many lifestyles that this is really one of many there's no reference wild type that's one thing and then the second thing is when you take an animal into a laboratory you know that it's not going to be getting the same

microbiome or the sun or the temperature variability or the prey or the predator or whatever it is but you you go you forego that degree of freedom and naturalism for an element of controllability and so in ecology they do a lot of mesoscale experiments where they'll set up like 200 fish tanks and add exactly three salmon and exactly 11 algae and set up these these tanks and they can demonstrate things like dynamic oscillatory systems or like a instability point with the ratios of different initial conditions of two different species or the role of different nutrients and then the the critique just like it is with animal behavior in the laboratory is like well that's not exactly the natural ecosystem and so it's just an interesting topic how in ecology there's this continuum super long scale systems like cosmology that you're never going to be able to uh intervene in or make experiments about like tectonic plate level understandings of biodiversity ranging to very local understandings that are often very controlled but then those are often subject to the same kinds of holistic criticisms that so many other frameworks are and so to turn again to active inference there's a way to build positively on this and just think great there's this mismatch between levels or at least apparently why don't we kind of think from the top down and set out a really big scaffold for multi-level systems and then we'll figure out how to map different levels and scales of analysis within this just like people learned how to use multi-level regression this is going to be like a variational tool at that kind of category or of that kind of type all right figure six so here maxwell on figure six i'll let you take a first pass but the next slide if you just you let me know if you want the box to come up but let's just focus on this uh image and maybe start off just like what does it convey or what is it there for but go ahead okay uh just let me pull up the the figure myself all right um it's called a particular partition it's a point yeah just to explain it i know that i just wanted in front of me while i explained well so basically this this figure um explains how uh you add i mean i i would i would prefer to explain things in terms of the figure figure 3 because this is the slightly more technically complicated uh version of the figure three like it figure six basically tells you how you uh i mean effectively take a bunch of states and like add add them to the superordinate scale but i think it's slightly easier to explain using figure three so what we basically do is we're going to take a bunch of states that comprise our system and then partition it in partition those states into markov blankets and then from there um construct a uh a reconstruct the system at the superordinate scale on the basis of these markov blankets so uh basically in new work um especially in the particular physics monograph uh we speak of this in terms of two operators that we use uh g and r uh g is a grouping operator effectively and it allows us to construct uh markov blankets from the base uh states uh that we're

considering and then r takes these mark all blankets and then reduces their dimensionality so i mean effectively we're going to be dropping some of the states uh to construct the states at the scale above so we start with a bunch of states and uh these are all the states that are of interest uh to us in the system uh so you know um these can i mean these can be literally any states whatsoever these techniques have been used recently to partition fmri data so when when you're dealing with fmri data each state is a voxel so a 3d pixel effectively and so what we do from there is you have all of your states and you have this continuous time series data typically and um so what you do is you construct an adjacency matrix um so an adjacency matrix is essentially you're plotting all of your states by all of your states such as the diagonal is basically like you know your degree of self coupling and what you do is you populate this matrix with uh coefficients that express how tightly uh this state and that state are coupled and then so from this adjacency matrix what you do is uh you take the jacobian and the hessian of these adjacency matrices so the jacobian and the hessian are basically the same matrix but rather than having the coupling coefficient uh in each entry what you have is the partial derivatives of each entry so effectively this is a technical and complicated way of saying what we're looking at is the relative rates of change of every variable with respect to every other variable the reason we're doing this is that a zero in these uh you know uh matrices that contain the partial derivatives means that effectively there's no relative change meaning that like you can vary one of the variables and the other one doesn't change so this is a way of testing for conditional independence is the short story so basically using these mathematical techniques what you do is you read off the conditional independencies from this coupling matrix this adjacency and from there you're able to rebuild uh well you're able to construct markov blankets in terms of the states that are independent of each other conditioned on other states so i mean to recap we started with a bunch of states that just composed the system that we're interested in explaining at one scale we put these states through a grouping operator that chunks it into markov blanketed particles based on the conditional and tendencies and so this takes us from the bottom of the figure to the top uh top left so what we're left with after applying the grouping operator is a bunch of markup blankets mark all blanketed particles so then in order to get to the next scale what we do is we reduce the dimensionality of this uh partitioned set of states so effectively what we're doing is we're dropping the internal states and the fast um stable modes of interaction at every scale in order to construct the states at the level above and this gets us uh basically from the uh it's the it's the other half of the image basically moving from these uh these partitioned uh states uh back to

uh a new set of states at the superordinate scale so conceptually what we're saying is that at a superordinate scale are literally constituted by the slow meta-stable modes of interaction between states at the level below so just conceptually this means that you take say an organ like the heart well the heart itself is constituted by the slower modes of interaction among heart cells so for example like you know uh the uh the main compartments of the heart for example are literally constituted by the modes of interaction uh between the cells that are components of those uh of those components effectively of those compartments um yeah so the then the reduction operator we're dropping the internal states because they're effective all of the information that you need to know about the internal states are already is already summarized in the blanket so you drop the internal states and you drop the fast stable modes of oscillation and that gets you from the uh the partitioned system back to uh a set of states at the next scale um so hopefully that explains what's going on but again the the that might sound very technical but the idea uh i think is pretty simple and it's just that we're constructing superordinate uh layers of the system from the slow modes of interaction of components at the subordinate scale very nice explanation maxwell really thanks a lot for that yeah super helpful so let's go over that because i think it's really critical and i think for some people it's like wait weren't they just talking about ecology so what just happened there how did we go from the big questions in ecology to this grouping operator loop that was just described and so the main pivot i think has to do with multi-scale systems and how we're going to be modeling across different scales and so what the question is is uh again thinking about understanding and modeling these dynamics of multi-scale systems and there's an unprincipled way and this is a principled not the only principled way but a principled way to do this multi-scale integration so an unprincipled way would be for example um you use method a to go from the fast dynamics to the next level so you have a heart cell simulator proprietary program heart cell sim 4 and then you make just another software pipeline that connects the outcome of that simulation to a model of the whole heart physiology and then you kind of run the the results from these two different models back and forth so maybe it works maybe it's functional so no one's saying it isn't it's just that the way that those two pieces were joined together was a bit a priori it was a little bit just chosen and it would be better if there was a way that we could for example take measurements across levels and use information that they provide to constitute our better understanding of the whole system so not just throw away the model that went um information that doesn't fit into each little bottleneck of information but how can we have this full variational in the um like constituting variance across multiple time scale as well as

variational from that physics side which brings along all these really positive benefits related to computability attractability and relationship with bayesian statistics and forney factor graphs and all these kind of cool areas it ends on bridging so if there's a hinge with multi-scale systems having a useful description then we're in the category of the physics models like spin graphs and game of life and then also at higher levels at the very least the tools for dealing with the kinds of things that those kinds of simulations like the game of life were invented to describe so that's sort of the main key point and this is what maxwell has run through and represented here is describing just like you could use model selection techniques to figure out which multi-level statistical regression model or which multiple anova was the most appropriate or whether to include an interaction term or not or a multi-um uh scale model from a bayesian whichever modeling framework you're working within hopefully there's a principled way to go about doing model selection and doing model comparison and so if i can just interject yes uh carl carl fristen um calls this the apart the particular partition and there's a there's a play on words here there's a double on the by particular partition he means partitioning the system into particles which are the internal states of a system plus the markov blanket but he also means it's a particular partition as in there are other ways of doing this uh there yeah there there's not one unique way of cutting things up uh and so the partition itself is named to reflect that fact i thought it was kind of fun to bring up yes now let me connect that point right there about one particular choice relating to dimensional reduction techniques and where uh degrees of freedom come into play for the experimenter so in real data sets you can do techniques like principal component analysis and you can do things like linear discriminant analysis which looks basically for group separation as an additive combination of principal components so it tries to explain the most variation with these orthogonal axes through the data that's principal components analysis now there's some data sets where the first principal component explains 80 and the next one explains five percent of the variance and has a long tail now whether you need the first principal component and you're just going to take only the first one or whether you take 99 of the variance being explained by 500 principal components where the explainability of principal component 74 on your netflix consumption doesn't really map to like a real world understandable trait it still can reduce your um uncertainty about a machine learning model but it doesn't like map onto something that's actual about the group anymore may be misaligned with all these other things those kinds of analyses are always up to the experimenter to decide how many dimensions they want to reduce to do you want to reduce down to a two-dimensional space just upon the data points in the principal component

one principle component two just remap onto the two dimensions that explain the most variants linearly or do you want to map into a 500 dimension space that's where a lot of the freedom comes into play so just like there's a particular principle component analysis is presented in a paper but principal component analysis is a general technique a particular system will have a particular decomposition statistically and in this way and that will also reflect degrees of or freedom in modeling it and in the statistics but the framework just like the anova framework goes a little bit beyond and isn't necessarily um about any specific system well one thing to point out is that there's a kind of recursivity at play i mean what you can actually see what is the best way to partition the system just kind of recursively in that like what you can do is take your system and then write right alternative generative models of how you think the system is actually behaving and then you can use free energy as a metric to see which out of these alternative models of the system's structure is the one that best explains its behavior so there's a meta-bayesian kind of aspect to this where like we're we're not only just like modeling uh the system as if it were an active inference agent but we can use generative models to assess uh which out of the possible ways of cutting up the states of carving up the system is the most appropriate and again it's using the same metrics uh so model evidence it's always model evidence i had a call with uh carl yesterday and he was saying well you know whatever the question the answer is always multiple evidence let me connect that's one more niche idea so we talked about the skilled intentionality framework and how that's part of you know the niche is the affordances and that can reflect a lot aspects that are learned or that are cultural but the niche can also include mercury that's influencing you and you can't taste it or detect it but it's changing your phenotype or something like that so again maybe some niche concepts are more affordance oriented like the ecological psychology niche is more oriented towards action unsurprisingly however for many ecologists when you talk about the niche that means like how much light is hitting the ground what temperature is it at night so that is also reflected by the realized versus the fundamental niche because especially when you're talking about those abiotic factors you're sometimes talking about oh yeah rocks that you can knock over but a lot of times again people just think of rocks as being in the physical niche so things can influence and not be reflected by sensory states and then all we have is the model so you have the model of decision making it doesn't know that it's being slowly drifting off into a non-adaptive state space because of the cup that has led into it all it is experiencing reducing its uncertainty is about is its generative model of its niche and that's more on the ecological psychology side and that can even be extended to like a molecular level like chemicals that

can't be tasted they can still influence you so it's not only about how the action is enabled in the world but that is the primacy when you're coming from an organism or a system of interest perspective and that's where you make this initial like what are the measurements to make like maxwell started off with you start with states those are measurements you could say we could measure this we could measure this thing over here we could do it over here but you got to start with at least the idea that measurements are going to be made or at least that there are certain distributions or types of measurements that could be made um one other thing with the grouping operator that you mentioned that kind of reminded me of click detection of conditional independencies and so the the other way of saying that is click detection of the conditional dependencies because in order to find out which ones are zeros you got to find out which ones are nonzero and then when you think about there's this representation ability to transform between the matrix and the network like through the adjacency matrix or more advanced relation related matrices then you can do click detection on a matrix so matrix is going to be able to make a network that has the exact same data in it whether it's just one zero like edges or not edges or whether it's a weighted edge or a different type of edge there's such an equivalence between matrices and networks that click on a social network or something like that is going to be also detectable on the matrix form so that's something to keep in mind when we're talking about like a matrix coming in here but then there's like maybe a network of interactions coming out another side okay any thoughts or questions that was definitely worth explaining maybe we could now return to six or we could also go to four where there's a few questions from previous i really wouldn't necessarily go sick honestly uh it's like the most technical slide like in in our papers sounds good so i mean like unless unless the really nitty nitty nitty gritty gritty gritty gritty detail is of interest i would just uh i would suggest that we not and i think figure three captures everything conceptually uh the figure six really just tells you how you add states to every particular thing so it's just running through the same thing but like okay we're taking this state and now what do we do with it uh so yeah we'll do 7.3 late night maybe if we need to but for now i think let's just return to figure four which we've actually seen in previous weeks and i just wrote down a few questions one was i believe alejandra's question from several weeks ago about what are some differences between simulating systems like those that can at least apparently engage in deep counterfactuals and morphological systems and so the question arose like okay but is the cell really doing a counterfactual analysis of alternate cell shapes it could be in a way that the brain may be doing some type of modeling of what behavioral states could be and then so that's the previous

question that we're definitely going to return to hear new perspectives on from anyone who wants to raise their hand and then the second set of questions is cool where does evolution fit into all of this and so a little bit more broadly how does this bayesian perspective play into evolution development and learning because this is a morphogenesis problem it's a devo problem and we know that eco and evo are going to come into play with devo so how do we think about all these levels and how does figure four make its appearance in this paper what what is it relevant what does it show so any thought i think that is just about how morphogenesis and in general higher order pattern formation happens under the variational framework uh so i mean we've discussed this in previous episodes but the idea is always essentially the same it's that uh so what is it to share a well in the context of active inference to share a generative model means that you share the same beliefs about the kinds of sensory consequences that follow you know the effects of states in the world and in particular the effects of my own actions in the world so the idea is that if we share a generative model we expect to perceive the world in the same way as a consequence of the same kinds of so uh pursue pursuing on that idea if we have the same generative model then if i just do my thing and you just do your thing we're eventually going to end up zeroing in on the same overall pattern because we share the same expectations about the kinds of things that we should expect conditions on the same kinds of so if if you have a system that's equipped with like state or yeah state dependent observation profiles so for example if i'm a heart cell i should expect to sense other heart cells and to you know register blood flow and whatever whereas if i'm a brain cell then i really shouldn't be registering blood or other heart cells at all so if you have these kinds of state-dependent observation uh distributions uh then you can start to see how sharing a generative model would allow for a bunch of components to zero in on uh well first of all on a target configuration and second of all kind of settle into a mutually coherent pattern of inference about what is my role in these higher order uh patterns that i'm part of um yeah and so the the idea there i guess is the so this is the hor the vertical stack idea that we discussed uh last last week uh this the variational ecology is a horizontal and a vertical story a vertical and a horizontal story it's a vertical story in that you have nested systems of systems of systems which is what i tend to emphasize in a lot of my work but at every basically in between every two scales there's a niche construction story where uh you know the the the body at large is a niche for the organs the organs at large are a niche for the cells and so on thanks i'm gonna address the evolution side and then return to this idea that maxwell was just talking about about shared generative models because that was also very helpful so

how does evolution play into this well let's imagine this morphogenesis creature that we're watching okay it turns out that there's a lot of parameters in this model that are detailed in the paper but it's not like all parameter combinations lead to the solution on the bottom right only specific priors and specific mapping relationships between sensory and action states will result in this exact kind of um light bulb shaped creature okay you can imagine there's some arrangement that might result in a circle there's some arrangement it might the state space is large and it might not fix upon a stable point or it might do it very rapidly what cuts through all of these models what does model selection evolution if it turns out that the wind is blowing from the north to the south in this image and that this shape it has worked in the past and it left more uh you know offspring that believed that that shape was the morphogenesis um position to approach you'll just see that around it doesn't mean it's the only solution it just means it is what has existed so this is sort of an ab initio or a computational simulation of an artificial situation obviously but in the realized world creatures are evolving then you find that only the organisms that have the phenotypes that adequately reflect regularities their niche are going to be left carrying forward and so it's kind of like wow how do the ants know to drag back this fly from 30 meters in the desert it's like it's all they do it's all they've been doing if they couldn't do it they'd be doing something else or they wouldn't exist it's not like you'd see an ant just struggling and failing to do it that would be a very short-lived strategy so there's wacky stuff that happens again when the niche is transiently misaligned with evolutionary priors or when an evolutionary prior tends towards just like accept everything into the nest you know pull in threads or something like that but i think that's really important to keep in mind that evolution often does model selection for us to actually find the solutions that work bottom right and the expectation action relationships top left that relate in morphogenesis top right through the extracellular target signal being the coordinating mechanism in this morphogenesis example on the bottom that's development mediated what appears to be by this basically monotonic decrease in free energy representing the utility of modeling this system in converting to or converging towards a developmental attractor using a free energy framework well notice that it spikes initially which is so all the agents are kind of trying to figure out what their position is and then they settle into like a mutually consistent uh group inference um and you know this work has been recapitulated across several scales uh ensen palacios and colleagues have a cool paper where they do this across three scales so they they have like a micro meso and a macro scale all premised on the same free energy functional and you get these nice layers of like organelles within organs within an cool and let me also go back to one

thing you said about the shared generative model so you talked about like the neurons sort of being in their niche you know wrapped in a specific type of glia or receiving certain types of communication at different developmental time periods from different inputs um there's almost this um not a complete replacement that you can do of having a shared generative model as you mentioned between the cells but it's they're part of a shared generative model at a higher level like through the neurons own expectations and its own niche it subsumes and it becomes a functional part of a larger generative model brains know things that neurons and so that is very naturally accommodated in a multi-scale framework because at the level of analyzing just the two cells you'd find that they act as if they had a shared generative model and so it would make sense oh at the level of the social interaction it's like two people are having shared generative models like they both know how the movie's going to play out or they both know whose turn it is to speak but then at a higher level and that's all happening at the same time it's also like being part of a shared that is the higher level whether that's thought of as collective computation or collective cognition or distributed intelligence whatever framework it ends up playing out as the brain knows things that the neuron doesn't and so the social group is having dynamics that constrain and facilitate the individual level in a way that specifically will evade the computation or the understanding of the individual so i think that's kind an interesting consequence of this and yeah good i mean yeah to address a question was asked initially about shared intentionality um i mean so we have this paper now out in entropy uh where we we basically argue that the free energy principle provides a formal semantics and by that we mean that so what it is really to act on a generative model is to have an intentional relation with uh some features of an environment that you're inferring constantly uh i mean that that's effectively what having a generative model allows you to do it allows you to go from your sensory data and from your architecture of priors and likelihoods to an estimation of what most likely caused your sensory data and you know intentionality kind of seems to come pre-packaged with that as in like you know if intentionality means being uh appropriately receptive uh or to you know features of an environment that stimulate you well then having a generative model just means being intentional in that sense and sharing a generative model it means that you you're participating in a form of shared intentionality so a shared way of systematically you know relating uh your observations and embodied priors to um like patterns of behavior and so on cool one other thought and this was something that our friend blue was talking about at our event a couple weeks ago was this multi-scale agency concept so the individual agency concept a relational edge agency then the small group and then the

community and so we can't necessarily directly experience but what can you tell somebody to participate in at the local level that does engender this higher order adaptive process to play out and so i think it's the space between having a shared generative model and being a shared generative model like we're going to have shared norms so that we're going to take cognitive diversity and help that map our search space for the critical problems that influence us all so we'll have the best search and we'll figure out the outcomes that are going to be reflecting this kind of a distribution that we want to converge toward that's the kind of well specified harnessing of differences within the niche because you you said maxwell um agents that see the world in the same way so i i know that you didn't mean that from the overlapping because literally no one's you know generative visual model is going to be the exact same but there's this question of how much seeing the same way versus basically the variation within the niche in the exact same way right of course not but it's it's a goldilocks situation because if everyone had exactly the same predictive model then we would never see literally differences in opinions or ideas and so when thinking about the heterodox or the novel or the informative the adaptive that space is really about the mapping of the individual you know uh action perception loops in a sense onto this collective level and there's uh individual states that can be healthy in one context and and unhealthy or considered unhealthy in another context and so it's just i think it's just the beginning of thinking about multi-scale systems from this way but then as we know people's concept about themselves and their social world changes their behavior and in ways that are really subtle kind of what is water so just very interesting stuff cool any other thoughts on this figure so just to go through the figures in reverse order we had six which will again it's a late night spooky special for another time five any thoughts on five also looks like there's a lot of uh equations on there not needing to unpack now four talked about three and then i mean five and six for those of you who really want to drill into the i mean we could have like a 7.3 or something at some point uh ideally with a few of the more like technical people to really unpack that because five and six like really unpack the math uh in some detail but i i don't know how useful it is for like a high-level conceptual discussion like we're having right now so perfect yep totally agree and sounds good about that yep well those were the figures does anyone else want to return to any slide or want to bring up any other thoughts or questions like what was something that was interesting that they heard about or something that they didn't expect for the discussion to introduce or another direction that they would like to see it go towards well early on there were some remarks about theory of mind and so on yeah so we have a whole paper on this that came out actually it

came out in behavioral and brain sciences so we're very proud of um it's called thinking through other minds and yeah it's it's basically like well it was a target article and there were like 27 or 28 tier commentaries and we wrote a response and so if you're interested in how this fits in with theory of mind and you know the the debates between simulation theory and theory theory and all that well we have like a paper length uh treatment of that um yeah and the i think the the kind of in a nutshell response is well you can you can both you can have your cake and eat it too using the variational framework i mean you might have noticed that this is sort of our argumentative strategy is often to say well there seems to be this split in the literature and actually both sides of the split can be made sense of and done justice too by appealing to this active inference framework and here as well i mean we we kind of strike a balance between these kind of explicit theory theory mind reading uh accounts and these more implicit kind of embodied resonance kind of approaches again by appealing to our multi-scale story right so i mean you know you have uh more explicit and more implicit forms of inference about others and uh the more explicit they become the more like the more properly you're heading into theory of mind territory um but i mean that that could also be a topic for discussion that'll in a later podcast uh but there's there's a whole paper length treatment though it was my point if it's of interest anyway super cool i'll check it out cool yeah i think that it will be really good for the next two months last two months of 2020 that's eight weeks so we'll have four papers the ones that we laid out we'll cover a few a few uh old stomping grounds and a few new areas as well for eight we'll be going into math a lot more and i think in some other social domains maybe with the scripts paper and yeah i'm just like looking forward to these conversations making our queue of papers for 2021 having special events if there's anyone who wants to arrange for like a special interview or a session modeling like a mod like a simulation approach these could all be kind of fun types of discussions to have because it's fun to bring in all these different angles on it hear from the medical and hear from the practitioners yeah awesome thank you so much again for the really nice uh treatment that this uh podcast gives to our work uh my gratitude is overflowing uh i hope you can good yeah these really is unpack and they help a lot and this is what the game is about just understanding and learning and turning it over hearing in different ways so thanks for participating i guess this is 7.2 but next time in 8 9 11 for the rest of 2020 we'll we'll finish out strong with more learning and more exploring areas that are new to everyone so send us any comments or questions that you have that we can unpack live the follow-up form is in the calendar invite so if anyone wants to provide any feedback there that'd be cool and just uh stay in touch this has

been a great team com so thanks so much everyone for participating very cool take care yep
peace